Modified HiTSKT: supervisor verification brief

19 October 2026 · Evidence baseline: **e1065c9** · Primary training seed: **42**

Decision: retain the three historical experiments as provisional evidence. Further verification is needed before presenting the complete metric sets as reproduced. No demonstrated training defect currently requires retraining. **Publication readiness: NOT VERIFIED.**

The model is **hierarchical HiTSKT with deterministic causal ELU+1 linear attention**. It is a modified model, **not genuine FAVOR+ Performer** and not an exact replication of every original-paper component.

Three test results: distinguish history from independent replay

Dataset	Historical CUDA AUC	Independent full CPU AUC	Masking / accepted data	Fresh CUDA replay	Next step
ASSIST2017	0.7096240159197755	0.7096236372142970	**PASS** / **PASS**	**NOT VERIFIED**	Verify further; no retraining yet
Junyi	0.7978262055191263	0.7978264739233538	**PASS** / **PASS**	**NOT VERIFIED**	Verify further; no retraining yet
EdNet-KT1	0.7693648364413692	0.7693647361509419	**PASS** / **PASS**	**NOT VERIFIED**	Verify further; no retraining yet

All three AUC differences pass the unchanged **absolute 1e-6** acceptance tolerance. **All three complete CPU metric sets FAIL that gate.** ASSIST fails accuracy, precision, recall, F1 and BCE; Junyi fails accuracy, precision, recall and F1; EdNet fails accuracy, recall and F1. These failures have not been relabelled as rounding or hidden behind AUC agreement.

CPU test coverage is **112,252** / **2,456,402** / **13,429,870 targets**, using the validation-selected checkpoints at epochs **40** / **32** / **30**, respectively. The historical CUDA prediction vectors and actual precision flags were not saved. Exact original CUDA-versus-CPU pointwise errors and classification-flip counts therefore remain unverified. Full float64 ASSIST inference still misses historical BCE by **2.13e-6**; tested CPU precision changes do not resolve it.

What the independent checks establish

Masking PASS: future-response and own-target perturbations, independent explicit causal attention outputs and gradients, target shifting, BOS/EOS, loss masking, student/batch isolation and session boundaries pass within the production collator's right-padding contract. Arbitrary externally supplied left/internal padding is outside the full-model guarantee. Sequence causality does not establish when an earlier response became available in real time.

Accepted-data reconstruction PASS: every accepted source and processed/store event was checked for mappings, sessions, student boundaries, chronological splits, label alignment and intended target coverage; no intended supervised target is duplicated or missing. These are exhaustive checks, not samples.

Dataset	Accepted source rows	Retained learners	Retained events	Sessions
ASSIST2017	942,807	1,388	885,335	12,402
Junyi	14,660,217	29,865	14,660,217	600,154
Full EdNet-KT1	95,293,926	116,548	81,940,867	2,577,988

ASSIST/Junyi inputs match the authors' distributed CSV bytes; their original publisher-to-author transformation remains **NOT VERIFIED**. EdNet checks cover all **784,309 student files**, ZIP CRCs and official archive byte identity. HiTSKT deliberately uses the first training session as context only.

Evaluation meaning and response-timing limits

The existing protocol is **chronological within-student, retrospective and teacher-forced**: fixed weights predict each target using earlier indexed responses, including earlier validation/test responses. It is **not student-disjoint**, a global-calendar holdout or a certified real-time presentation-time evaluation. An earlier event index does not prove that its answer or correctness feedback had arrived.

Dataset	Test-set timing observation	Scientific implication
ASSIST2017	7,059 targets have an exposed earlier recorded end after current presentation.	Problem/step completion does not establish correctness release. Actual availability NOT VERIFIED.
Junyi	2,165,594 targets have a tied-start prior label; all source rows have start=end.	Zero recorded overlaps cannot certify instantaneous feedback. Actual availability NOT VERIFIED.
EdNet-KT1	4,089,385 targets have a prior completion-proxy overlap; 3,364,799 expose a prior same-episode label.	KT1 lacks final submission/feedback events. Pre-bundle prediction claims are not established.

These observations are **not counts of confirmed leakage** or estimates of metric inflation. Response-release timing needs separate evidence. Do not silently substitute availability-aware V2 or implement it as part of this report.

Baseline comparison: historical scores, not fresh reproductions

All entries below are **historical stored test AUCs**, rounded to nine decimals. **None of the 11 completed baseline metric sets has an independent fresh replay in these audits.** Checkpoint hashes and validation selection pass artifact checks; this is not metric reproduction.

Dataset	DKT	DKVMN	SAKT	RKT
ASSIST2017	0.706999742	0.682252702	0.738465565	0.748626490
Junyi	0.747684150	0.741578462	0.788796178	0.791254849
EdNet-KT1	MISSING	0.674105992	0.763007857	0.747547020

Recorded HiTSKT AUC is below the strongest ASSIST baseline and above the best available Junyi/EdNet baselines by **0.006571357** / **0.006356979**. These are single-seed descriptive differences, **not demonstrated significant gains**. Features, history budgets and training target populations differ. Correct saved history windows are DKT/DKVMN **199**, SAKT **99**, RKT **49**; this corrects a descriptive error in the earlier report. RKT raw relation counts still need an independent reconstruction. DKT/EdNet has no final result; its last optimizer contains **21,749 positive infinities**. Do not resume/reset it.

Minimum remaining verification, in priority order

1. **Close the HiTSKT metric discrepancy:** on a genuinely idle independent GPU, replay all three unchanged best checkpoints with original batches and settings. Save runtime flags, hashes, local predictions and all seven metrics; apply the original 1e-6 gate and report failures without retuning tolerance.
2. **Verify comparisons:** independently replay completed baseline checkpoints, prioritizing RKT/ASSIST, RKT/Junyi and SAKT/EdNet; independently reconstruct RKT's training-only relation counts. Keep missing DKT/EdNet missing.
3. **Freeze defensible claims:** retain the retrospective within-student scope and disclose timing/provenance limits. Real-time, student-disjoint, FAVOR+ or statistically significant-superiority claims need separate evidence. V2, FAVOR+ and additional training require an explicit research decision.

The only visible H100 MIG remains occupied by protected training; **no CUDA replay was attempted**. On 19 October, **59 relevant controls passed again** and all full cached CPU metrics recalculated identically. Original artifacts and prior evidence remain preserved. See the verification status and replay plan for hashes, acceptance rules, detailed scopes and the resource gate.

Evidence: numerical diagnosis, timing analysis, research decisions.